

High performance Analog and Mixed signal VLSI System design

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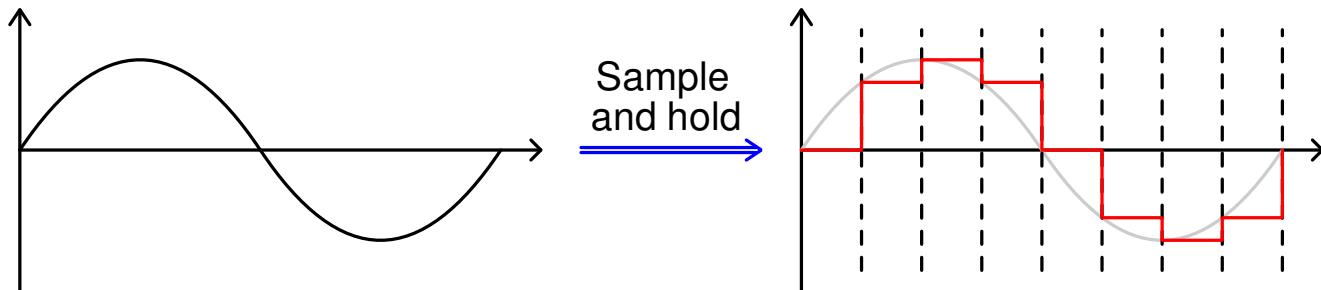
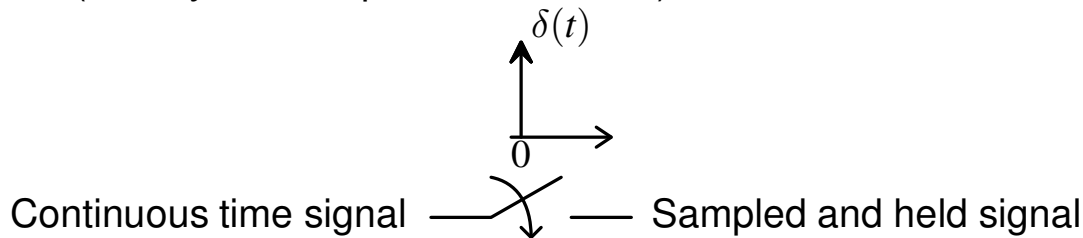


December 14, 2021

Sample and Hold circuits

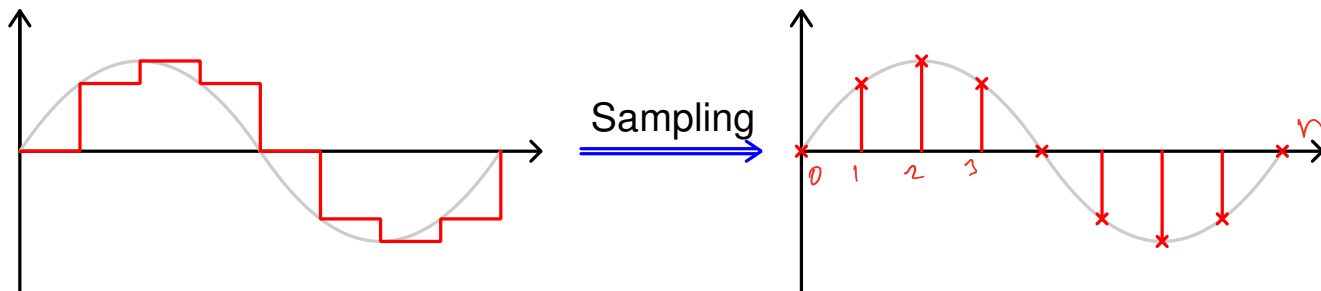
Sampling operation

- Sampling operation involves extracting the values of an input signal at specific instants of time (usually done at periodic intervals)



Sampling operation

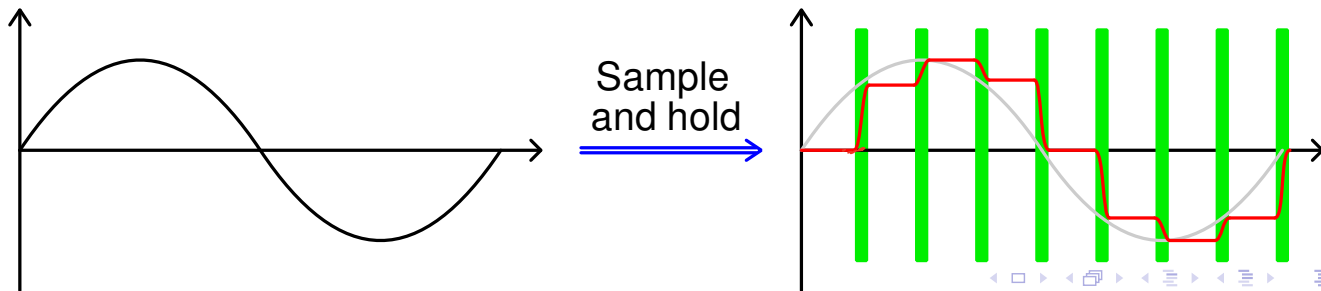
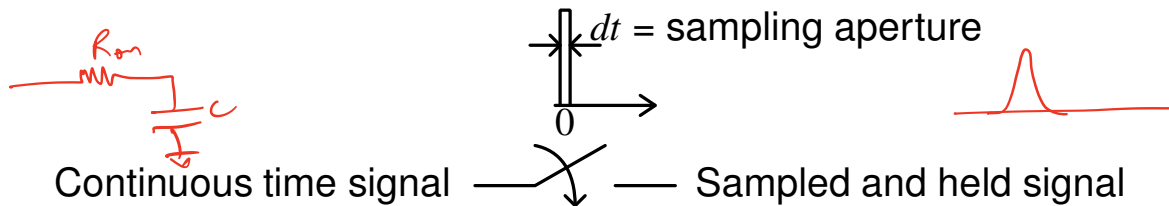
- Conversion from continuous time to discrete time signal usually involves 2 steps
sample and hold $\rightarrow$ discretize.



- Ideal sampling operation with an impulse signal requires infinite bandwidth and is not practical.

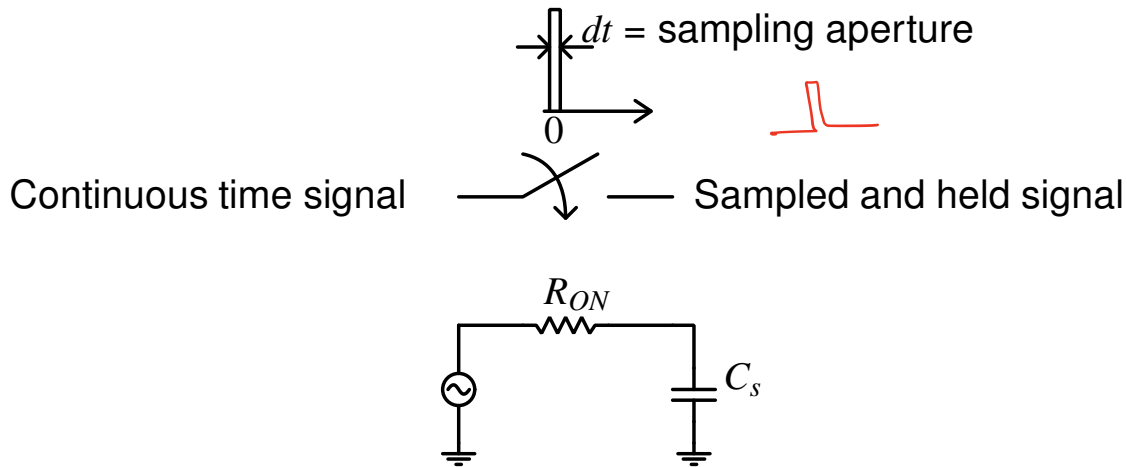
Sampling aperture

- Sampling aperture is the time duration over which the input is sampled. For ideal sampling with impulse signals the aperture is zero.
- Practical sampling takes the average of the input signal over a small period of time, which is the aperture time of the sampling operation.



Sampling bandwidth

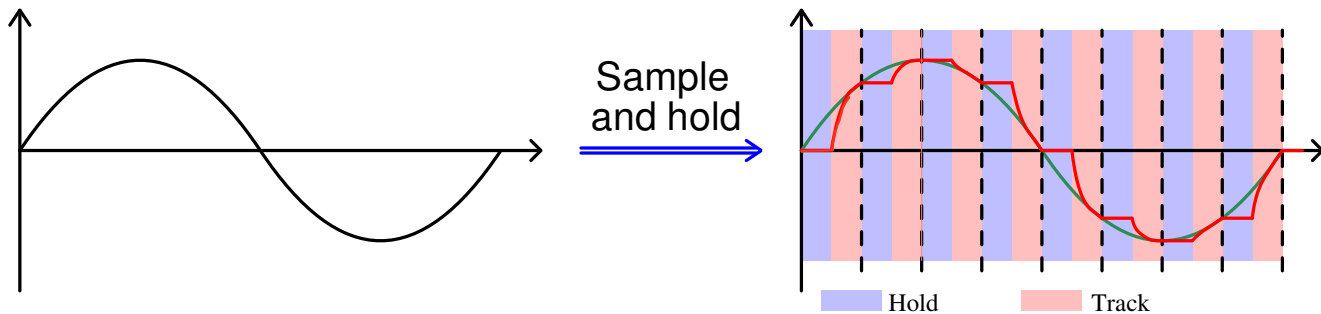
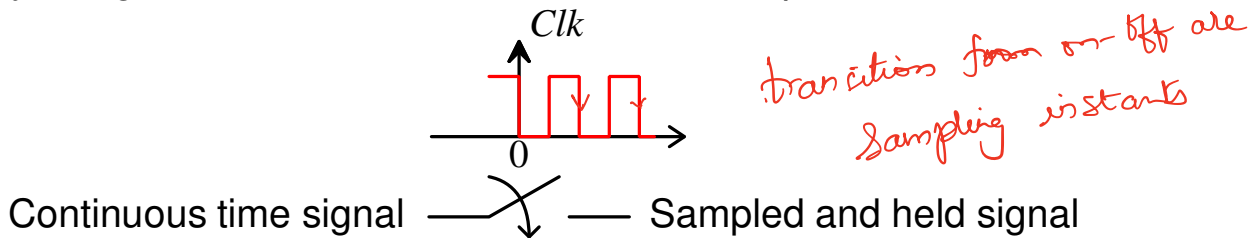
- Due to the finite on resistance of practical sampling switches, the sampling bandwidth is finite. This results in an error in the sampled value.



- When the sampling switch is on during the aperture time, the R_{ON} of the switch and sampling capacitance form a low pass filter, which determines the sampling bandwidth. This limits the maximum frequency of signal that can be sampled.

Track and Hold based sampling

- The constraints on R_{ON} put by sampling aperture time and sampling bandwidth can be relaxed by using track and hold circuit, instead of sample and hold.

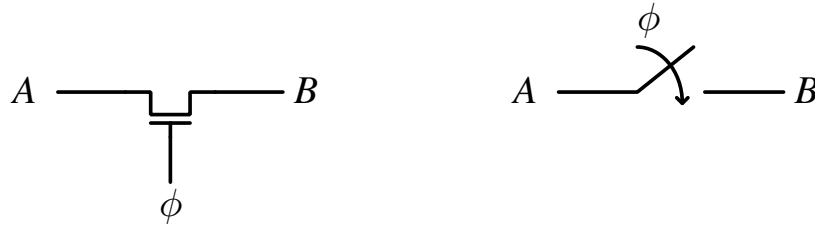


Track and hold based sampling

- Sampling bandwidth $\approx$ bandwidth of circuit in track mode.
- Sampling aperture depends on how fast the switch turns off and on the rise time of the clock signal.
- Noise and Jitter in the sampling clock also introduces an error in the sampled values.

MOSFET as a switch

- An ideal switch has zero on resistance and infinite off resistance.
- MOSFET can be used as a switch if we swing between cutoff and linear regions.



- When ϕ is high (i.e. Logic 1 in the case of ideal switch, $> V_{tn}$ in the case of NMOS as a switch) the switch is on and there is a low resistance path from A to B .
- In the case of MOS switch, we can calculate the on resistance using the drain current equations in the linear region.

On resistance of MOS switch

- The current through a MOSFET in linear region is given by

$$I_{DS}^{lin} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{tn}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

- For small values of V_{DS} the term $\frac{V_{DS}^2}{2}$ can be neglected.

$$I_{DS}^{lin} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{tn}) V_{DS}$$

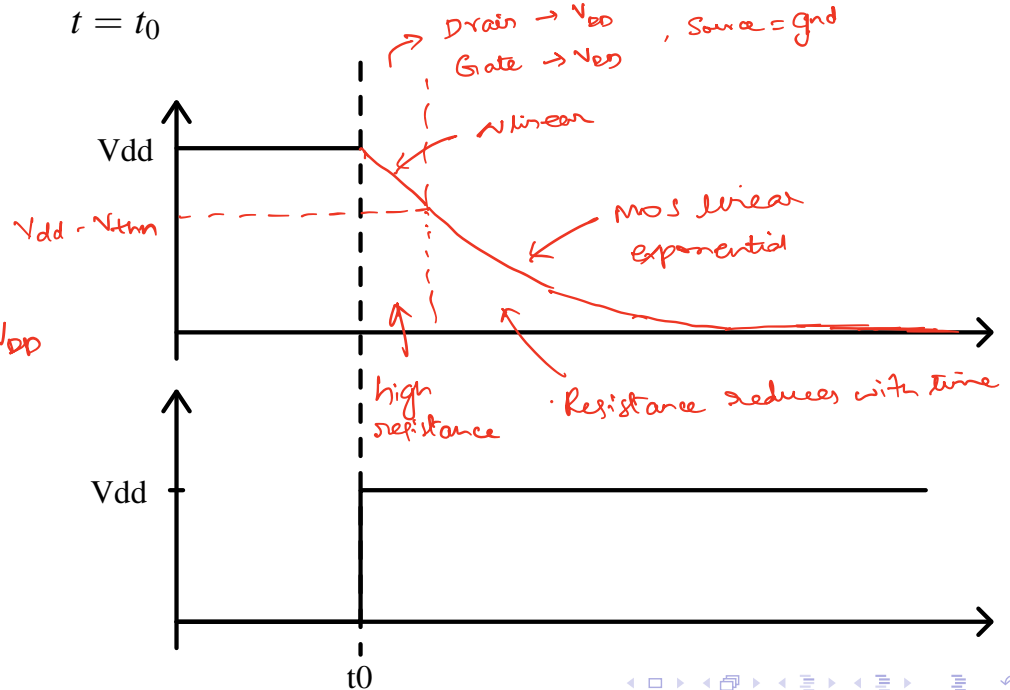
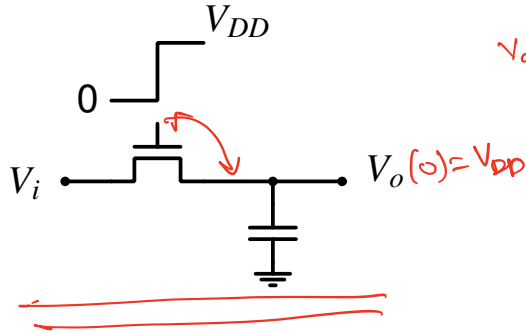
- The resistance in the linear region can then be found as

$$R_{ON} = \left(\frac{\partial I_{DS}}{\partial V_{DS}} \right)^{-1} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{tn})}$$

- This shows that the ON resistance is constant. However, this is true only as long as our approximation if $\frac{V_{DS}^2}{2}$ being negligible is valid.

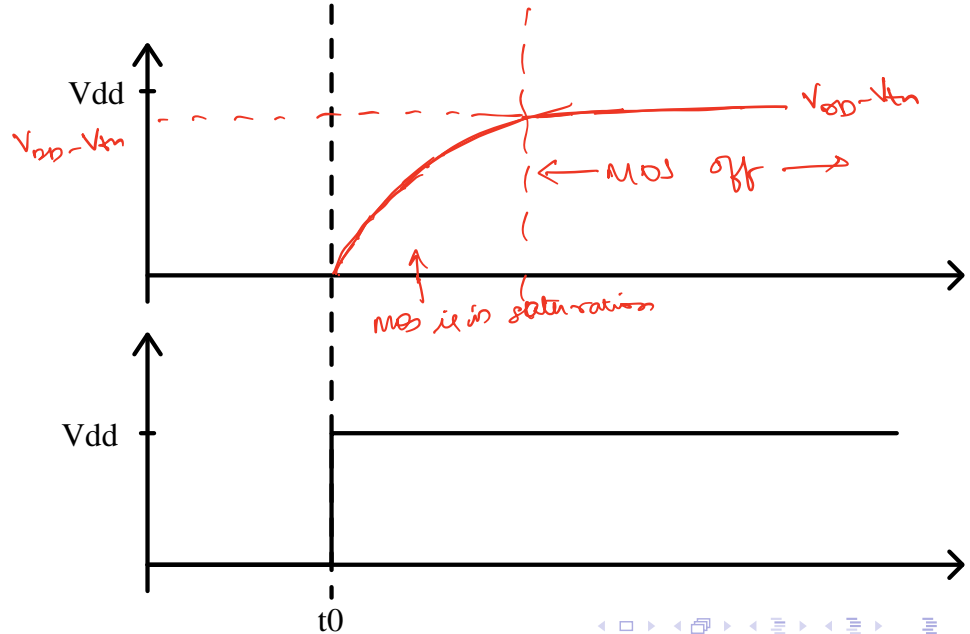
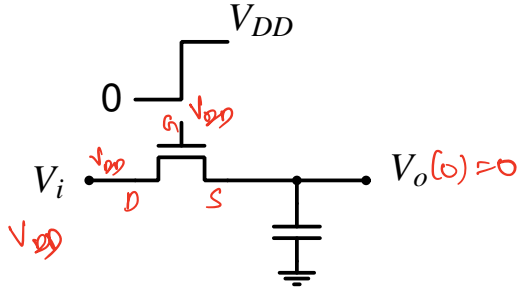
MOSFET as a switch - NMOS

- If $V_o(0) = V_{DD}$ and $V_i = 0$ and the switch is turned on at $t = t_0$



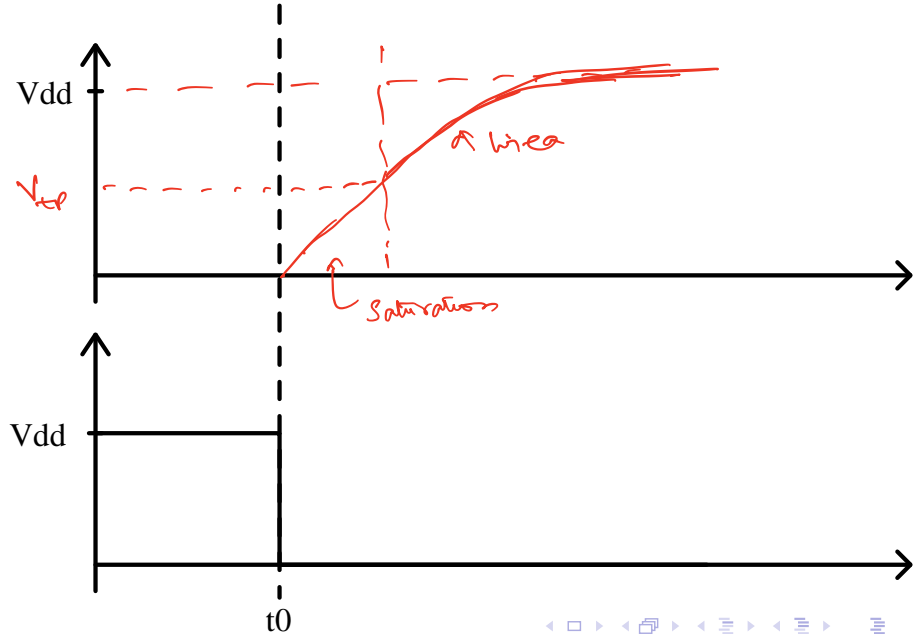
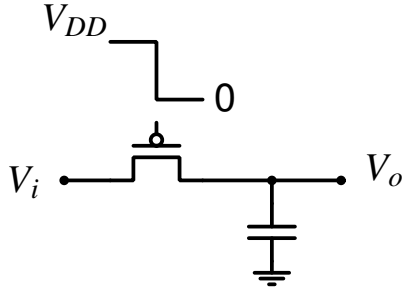
MOSFET as a switch - NMOS

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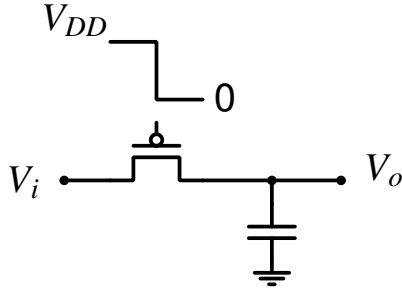
MOSFET as a switch - PMOS

- If $V_o(0) = 0$ and $V_i = V_{DD}$ and the switch is turned on at $t = t_0$

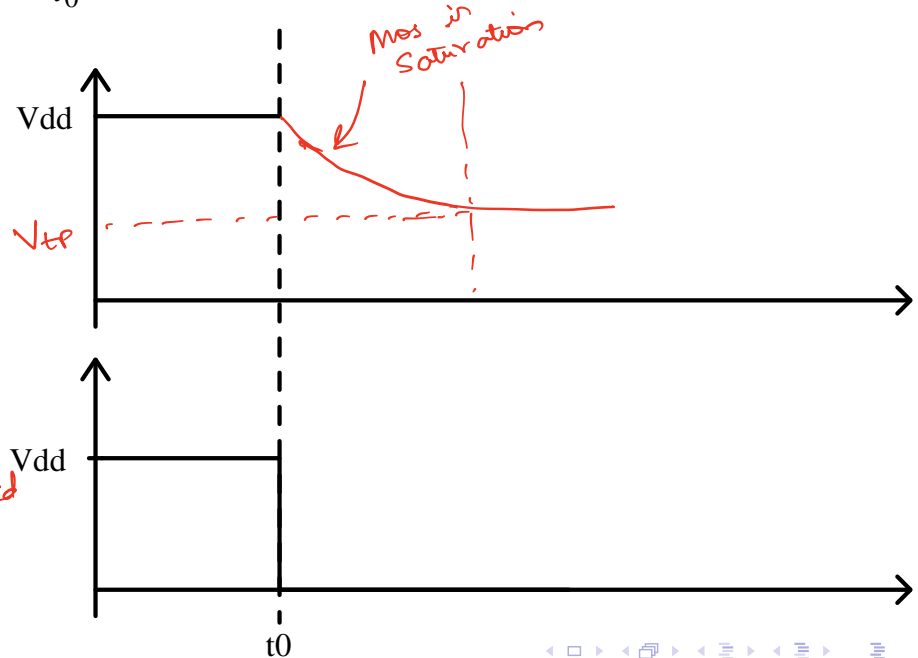


MOSFET as a switch - PMOS

- If $V_o(0) = V_{DD}$ and $V_i = 0$ and the switch is turned on at $t = t_0$

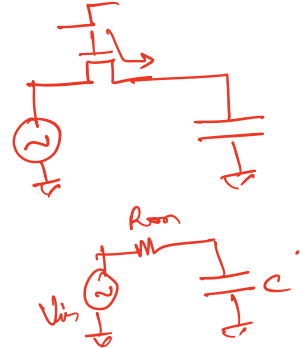
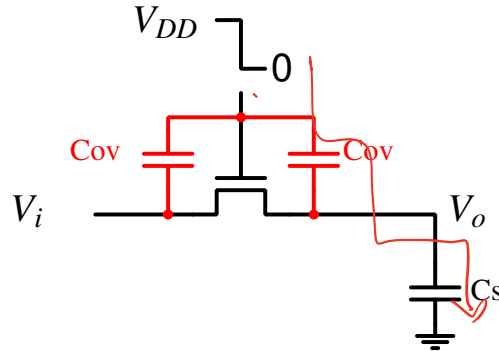


PMOS good for passing voltage closed to V_{DD}



Clock feedthrough

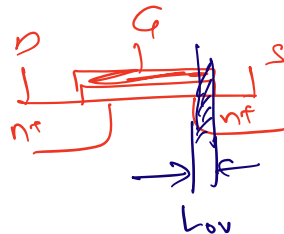
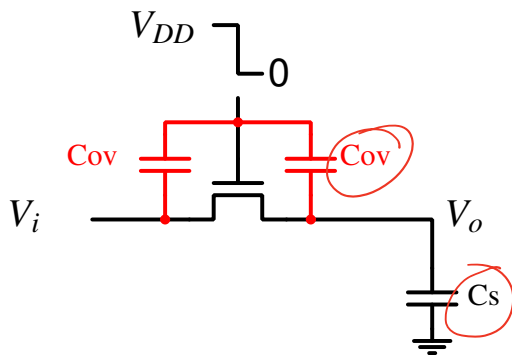
- Parasitic gate to drain and gate to source capacitance will result in coupling of clock signal to the output capacitor.



- Only the on to off clock edge affects circuit performance.

Clock feedthrough

- Parasitic gate to drain and gate to source capacitance will result in coupling of clock signal to the output capacitor.



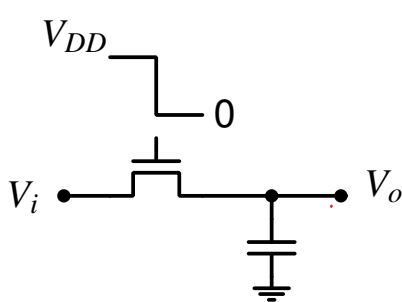
- Only the on to off clock edge affects circuit performance.
- The error in the sampled voltage due to clock feedthrough is given by

$$\Delta V = \frac{C_{ov}}{C_{ov} + C_s} \rightarrow \text{error is constant (independent of } V_{in})$$

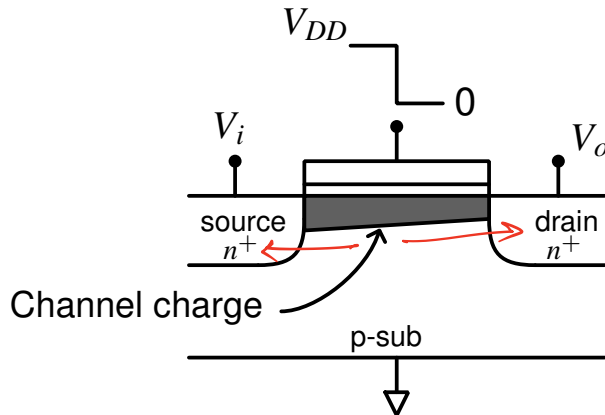
- $C_{ov} = C_{ox}L_{ov}W$

Charge injection

- When the MOS switch turns on, an inversion layer is formed by drawing charge from the substrate, source and drain. Consider an example when $V_i < V_{DD} - V_{Tn}$.



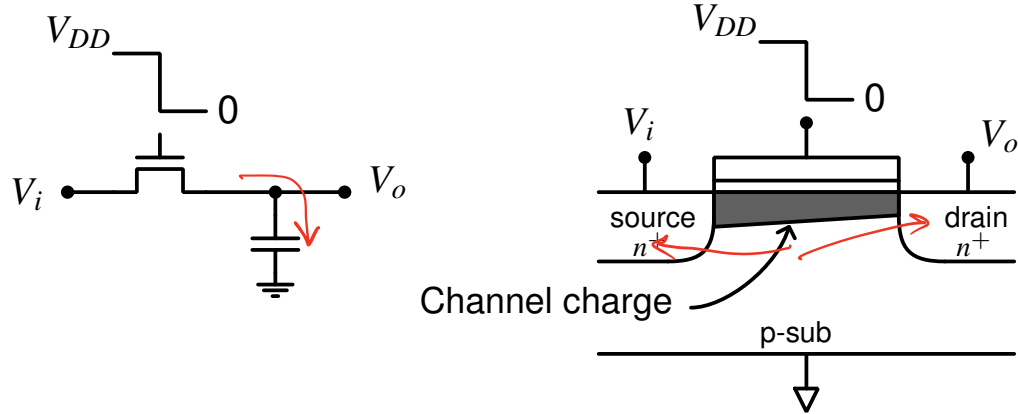
when switch is on
 $V_o = V_i$



- The channel charge is given by

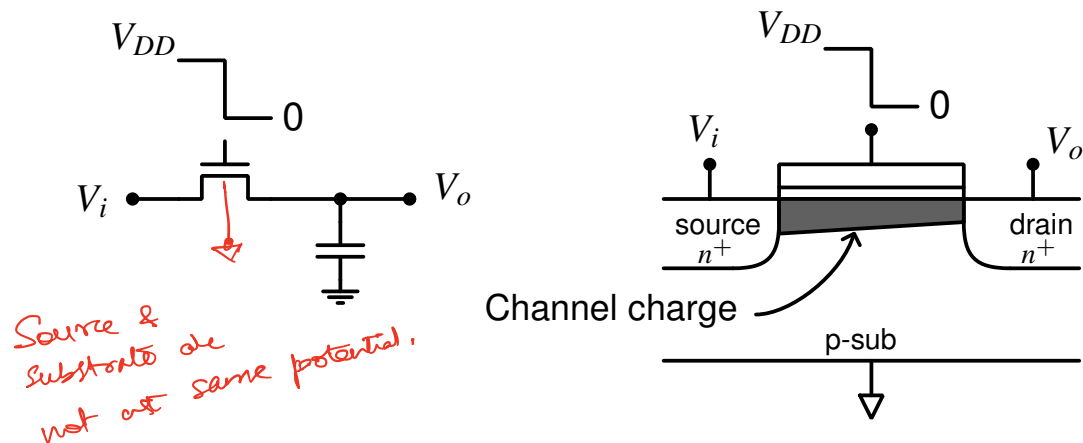
$$Q_{ch} = WLC_{ox}(V_{DD} - V_i - V_{Tn})$$

Charge injection



- What happens to this charge when the switch turns off? It has only two paths to ground via the source and drain terminals.
- A reasonable assumption is that 50% of the charge goes to the output node and 50% goes to the input node. (Note that these two nodes are at the same potential just before the switch turns off).

Charge injection



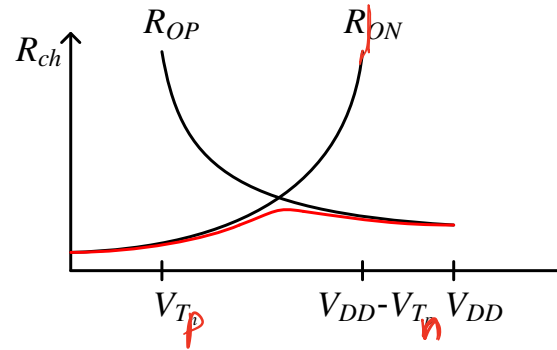
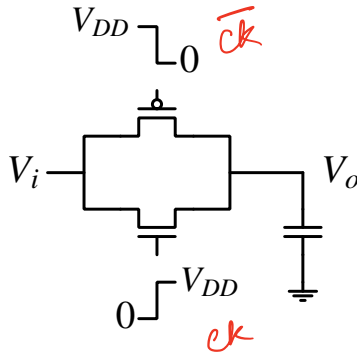
- The error in the sampled output is approximately

$$\Delta V_o = \frac{Q_{ch}/2}{C_s} \rightarrow \text{error depends on } V_i$$

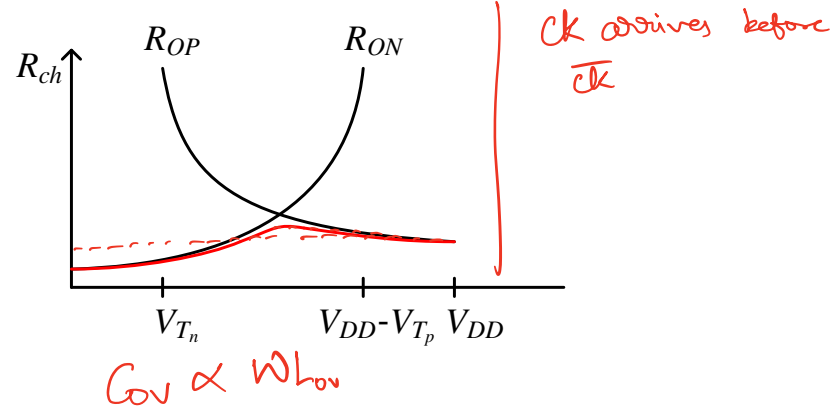
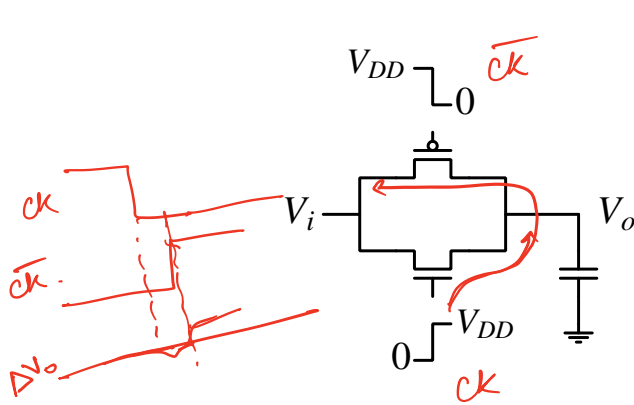
- The error depends on the value of V_i , which makes the sampling process non-linear.
- In this simplified analysis we have not considered the substrate bias effect on V_{Tn} of the sampling switch which will only make matters worse.

Transmission gate as a switch - a (somewhat) better sampling switch

- Using a complementary pair of NMOS and PMOS transistors, we can make a better switch which has reduced effects of charge injection and clock feedthrough.
- The PMOS transistor passes voltages close to V_{DD} well and the NMOS transistor passes voltages close to 0 well.

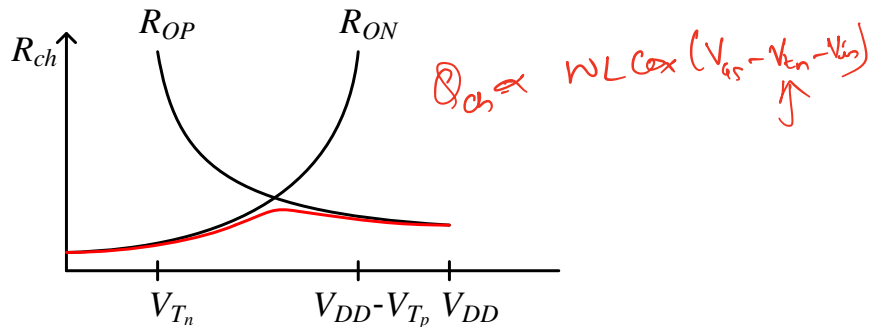
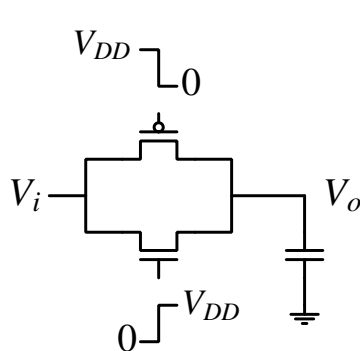


Transmission gate as a switch - a (somewhat) better sampling switch



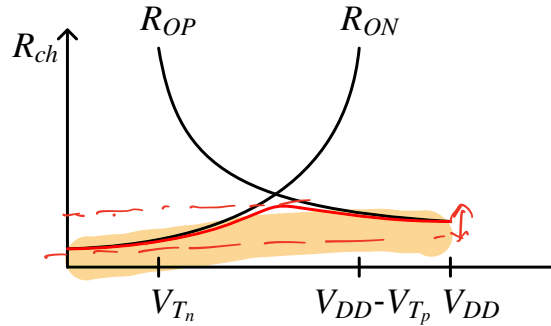
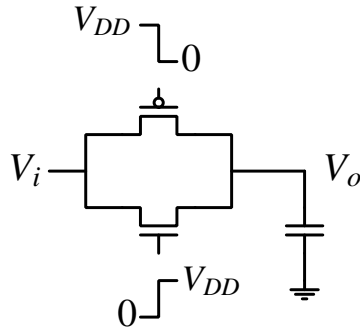
- Since the clock signals are complementary, the clock feedthrough will have opposite polarity.
- The error due to clock feedthrough will be minimized if the width of the NMOS and PMOS are equal.
- This needs differential clock signals which are well matched (matched in time of arrival).

Transmission gate as a switch - a (somewhat) better sampling switch



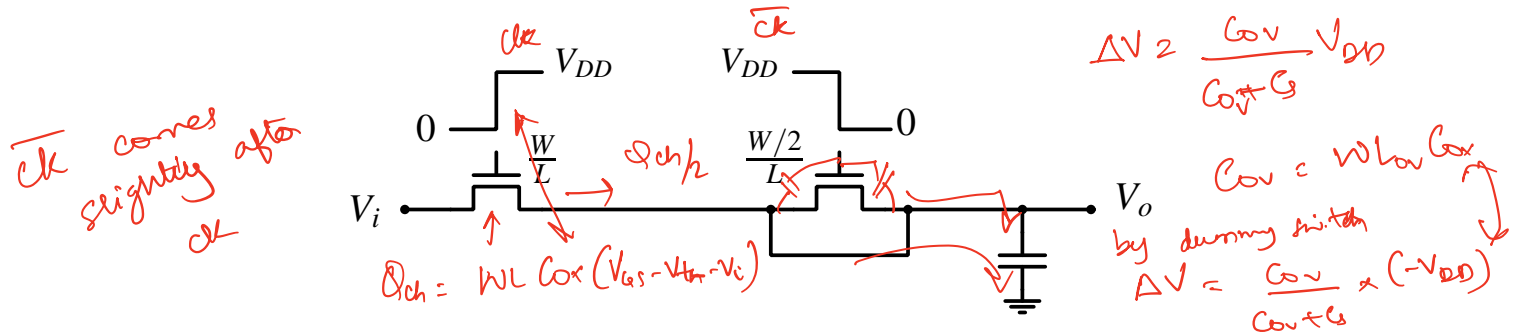
- Since the transistors are complementary, the channel charge is also complementary → electrons for the NMOS and holes for the PMOS transistors.
- Hence, the charge injection during the turn-off transition will cancel each other at the output.
- The error due to charge injection will be minimized if the geometry of the NMOS and PMOS transistors are equal (assuming $V_{Tn} = V_{Tp}$).

Some problems with Transmission gate switches



- It is often difficult to exactly match the arrival times of the complementary clocks with mismatches during fabrication.
- The variation in the switch resistance over the dynamic range introduces non-linearity in the sampled output. This often limits the usability of this type of sampling switch.

Reducing charge injection and clock feedthrough by using dummy transistors

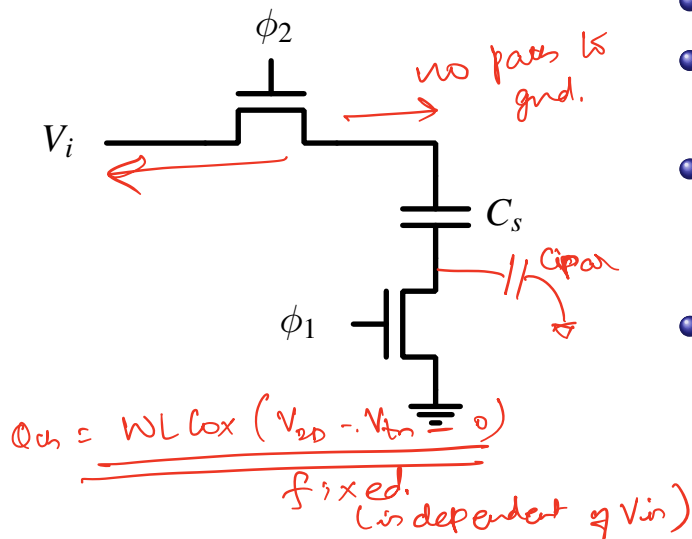


- The dummy transistor is of half the width and driven by a clock that is out of phase with the sampling clock.
- The charge injected at the turn of transient is used up in forming the channel in the dummy transistor.
- The error due to clock feedthrough is cancelled by the feedthrough from the dummy transistor which will be of the opposite polarity.

Bottom plate sampling

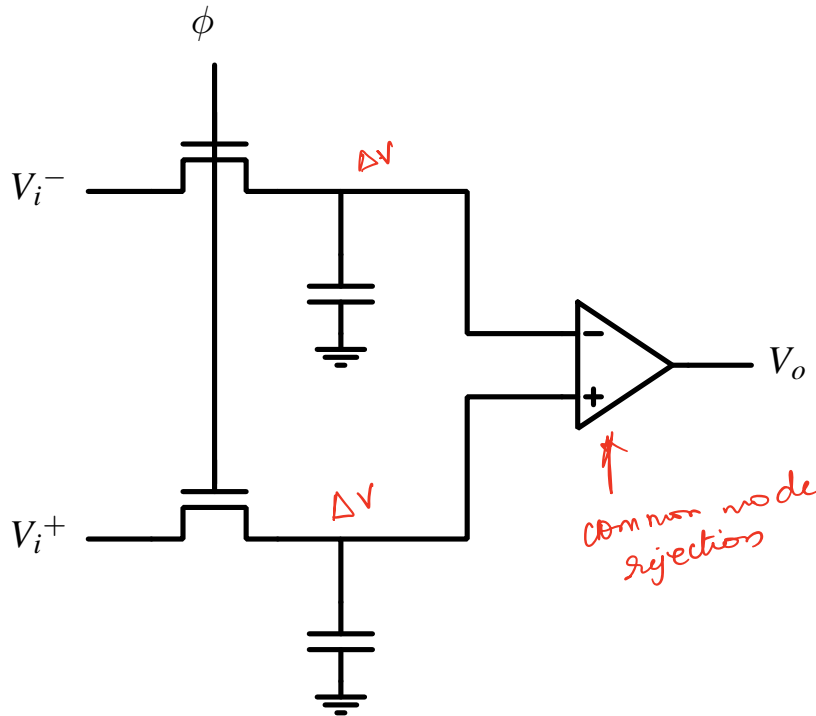
ϕ_1 & ϕ_2 are on for sampling -
(tracking)

- Charge injection error can be eliminated using what is known as bottom plate sampling.



- ϕ_1 should turn off slightly before ϕ_2 .
- The charge injection by M_1 will be constant and result in a simple offset error.
- The charge injected by M_2 will have to go to the input side as there is no path to ground on the output side.
- In practice, small parasitic capacitance from output side to ground will result in small error due to charge injection.

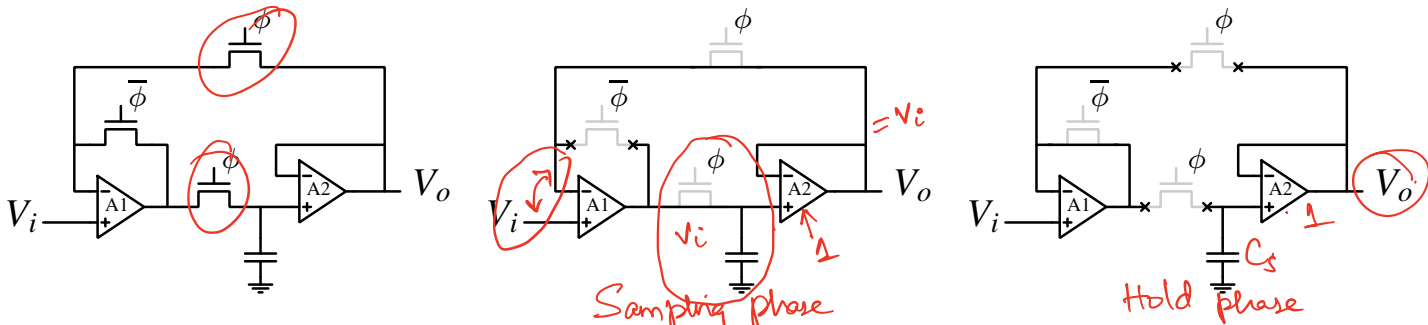
Differential signals sampling



- Clock feedthrough and charge injection appear as common mode signals to the differential amplifier.
- Charge injection is cancelled to the first order.
- Residual error due to the dependence of the injected charge on the input voltage will remain.
- This can be combined with bottom plate sampling to reduce the offset and error due to charge injection.

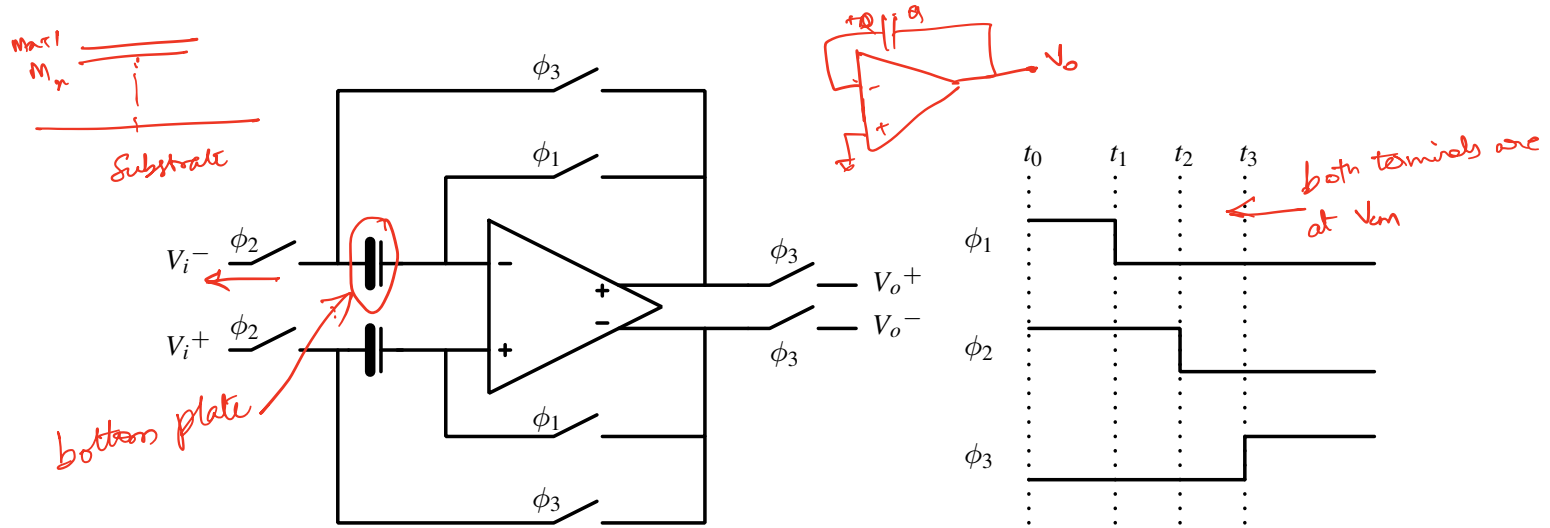
Buffering the input

- Many times the input being sampled may not be able to drive the sampling capacitor value. In such cases it is required to buffer the input before sampling.



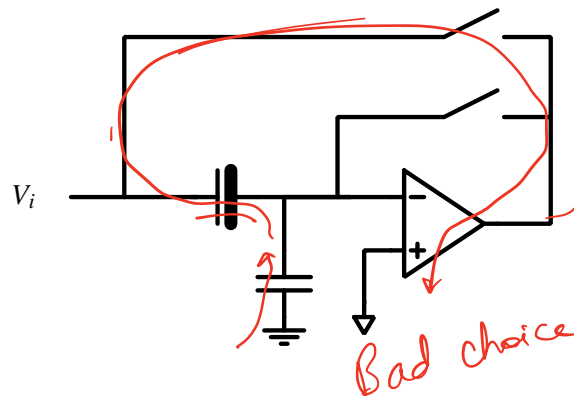
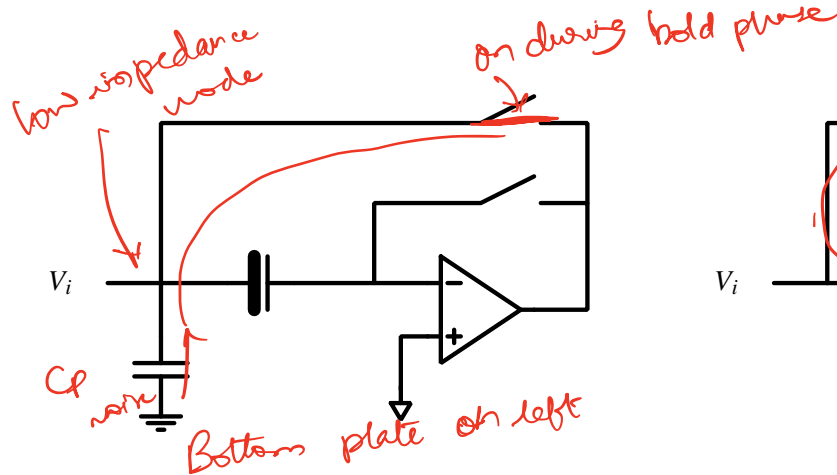
- This circuit still suffers from charge injection. We can use some of the compensation techniques discussed earlier in this circuit as well.

Fully differential sample and hold



- The thick plate of the capacitor should be the bottom plate (physically at the bottom in the metal stack).
- This reduces the error in the sampled value due to coupling from the substrate.

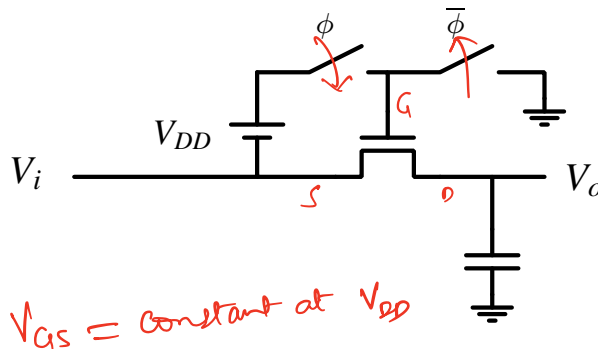
Substrate noise coupling to sampled voltage



- It is better to connect the node susceptible to noise to a low impedance node in the circuit.
- *Note that this is different from the bottom plate sampling discussed earlier.*

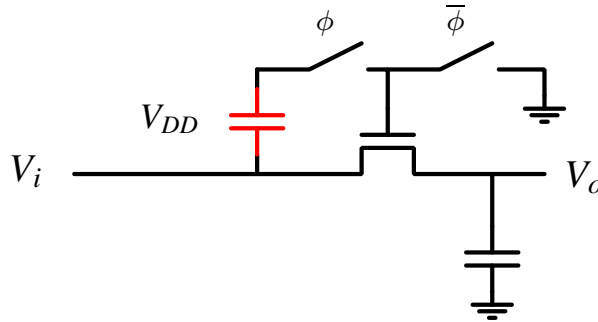
Bootstrapped sampling

- All the sampling circuits discussed suffer from distortion due to input dependent resistance of the sampling switch.
- Bootstrapped sampling is a technique where the switch resistance is kept constant by maintaining a constant V_{GS} across the sampling switch.



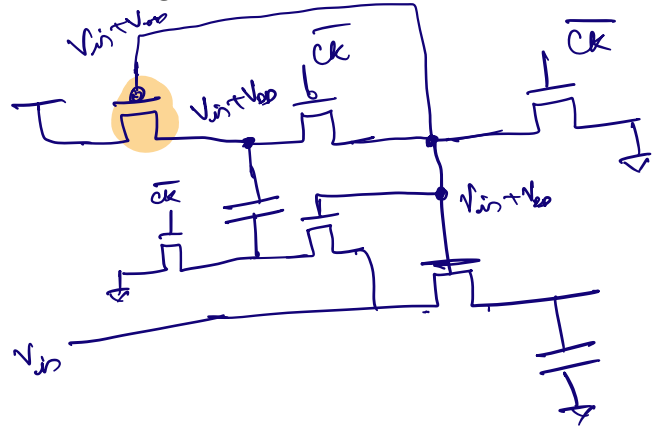
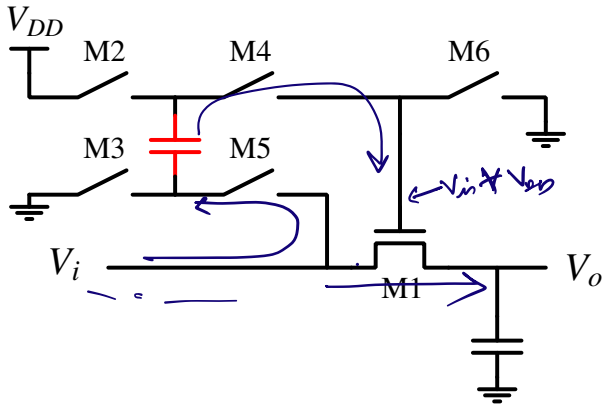
Bootstrapped sampling

- The battery can be replaced with a precharged capacitor (charged to V_{DD}) for implementing the bootstrapping.



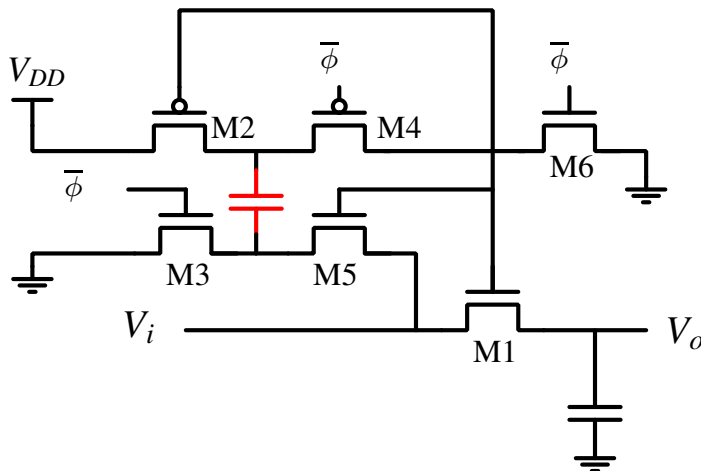
Bootstrapped sampling

- The complete scheme can be implemented using switches as shown below



Bootstrapped sampling - transistor level circuit

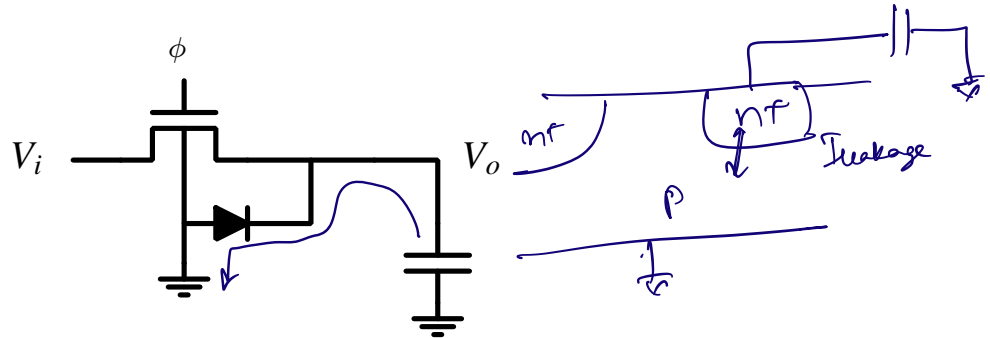
- The complete circuit realization using transistors is as shown below



- Because of the fact that the overdrive of the transistor is maintained constant, the distortion is very low.

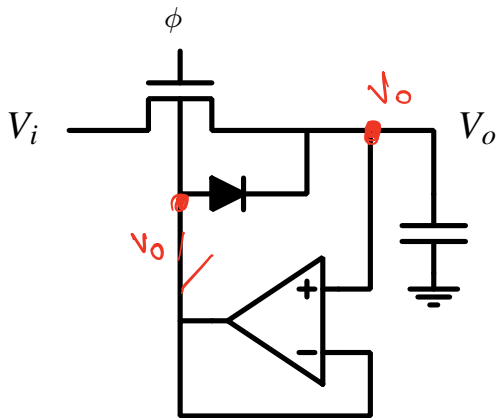
Leakage current in sample and hold circuits

- We have investigated many sampling circuits and distortion in sampling operation.
- After the sample is taken, leakage currents in the sampling switch will result in error in the sampled value which is of a concern if long hold times are desired.
- One of the main leakage paths is through the parasitic diode between drain and substrate as shown.



Leakage current in sample and hold circuits

- Leakage can be reduced by using triple well transistors and driving the substrate to the sampled voltage.
- One of the main leakage paths is through the parasitic diode between drain and substrate as shown.



- As both terminals of the diode are held at the same potential, the leakage current through the diode is zero.

Questions and Discussions

References

- ① Textbook: Paul Gray, Paul Hurst, Stephen Lewis, Robert Meyer, "Analysis and Design of Analog Integrated circuits"
- ② Textbook: R. Jacob Baker, "CMOS circuit design, layout and simulation"
- ③ B. Razavi, "The Bootstrapped Switch [A Circuit for All Seasons]," in IEEE Solid-State Circuits Magazine, vol. 7, no. 3, pp. 12-15, Summer 2015, doi: 10.1109/MSSC.2015.2449714.

Further reading

- ① M. O'Halloran and R. Sarpeshkar, "A 10-nW 12-bit accurate analog storage cell with 10-aA leakage," in IEEE Journal of Solid-State Circuits, vol. 39, no. 11, pp. 1985-1996, Nov. 2004, doi: 10.1109/JSSC.2004.835817.